Innocent Kisoka

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Machine Learning — generative models, careful evaluation, SciML, HPC

PROFILE

Machine learning researcher (M.Sc. Artificial Intelligence, USI, Switzerland — graduated 26 September 2026; B.Sc. Computer Engineering, Kyungdong University, South Korea). I build and evaluate learning systems carefully: LoRA/PEFT adaptation for language models, multi-seed metrics, structure-visible SciML, and CUDA/HPC experiment software. Evaluation comes first — when scores look fine but behaviour is off, I redesign the checks before changing the model. Interested in inventing ML methods that hold up under real deployment pressure, not only benchmark gains. Comfortable moving from small reproducible experiments to multi-GPU runs.

TECHNICAL SKILLS

ML / generative AI	PyTorch, LoRA/PEFT, preference-style training, multi-seed evaluation, Transformers
SciML / structure	FEM+PINN scientific ML, GNNs (PyG), topology-aware overfitting checks
Systems / HPC	CUDA C++, multi-GPU CSCS pipelines, MPI+CUDA, reproducible experiments
Core	Python, Git, Linux, scientific writing (English). Generative AI with LLMs (skills)
Languages	English (C1) • Italian (basic) • Korean (B2) • Swahili (native)

EDUCATION

M.Sc. Artificial Intelligence Sep 2024 – 26 Sep 2026

Università della Svizzera italiana (USI), Lugano, Switzerland — officially graduated 26 September 2026

- Coursework: Deep Learning, Scientific Machine Learning, HPC, Advanced Topics in ML, Computer Vision, NLP.
- Thesis: multilingual LM preference optimisation — LoRA/PEFT, multi-seed metrics, multi-GPU at CSCS.

CSCS–USI Summer School: Effective HPC & Data Analytics Jul 2026

Swiss National Supercomputing Centre (CSCS), Switzerland — CUDA, GPU architectures, MPI+CUDA.

B.Sc. Computer Engineering Sep 2020 – Aug 2024

Kyungdong University, South Korea.

PROJECTS

LLM adaptation & evaluation (Master's thesis) 2024 – 2026

- Closed-loop evaluation caught behaviour failures when automatic scores looked fine.
- Multi-seed LoRA/PEFT preference-optimisation experiments on multi-GPU pipelines at CSCS.

Cardiac diffusion PDE — FEM + PINN (SciML) 2025 – 2026

- Couples FEM structure with PINN-style learning so the model stays physics-visible rather than a black box.

CUDA / HPC numerical software 2025 – 2026

- Kernels, stencils, async pipelines, PDE mini-apps — experiment software for serious numerical runs.

GNN replication — topological overfitting 2025 – 2026

- Reproduced structure-aware checks in PyG and documented topology-driven overfitting.

EXPERIENCE

Teaching Assistant — Web Development Feb – Jun 2023

Kyungdong University, South Korea

- Supported labs and mentoring for undergraduate web-development coursework.

AWARDS

2024 — Scholarship, Foundation for the Lugano Faculties of USI • 2022 — Programming Excellence Award, Kyungdong University • 2020 — KDU Global–IUCEA 100% Scholarship